10Pearls Shine's Internship Report

AQI Predictor Model for Karachi Using ML

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Introduction:

Air pollution, causing 7 million premature deaths annually. The Air Quality Index (AQI) is crucial for translating complex air data into numerical scale allowing public to quickly understand risks, and take protective measures.

An application that predicts AQI can help people to understand the air quality for upcoming days so that they can prepare for it.

The AQI Predictor Model is a machine learning based model for Karachi to detect air quality for next 3-days. Three model were trained on real-time data of Karachi for predicting the AQI.

Data Collection:

The dataset for this project was collected from both historical records and real-time sources to predict AQI in Karachi. The data has pollutant concentrations like PM2.5, PM10, CO, NO_2, O_3, SO_2, and NH_3. The models were trained on both datasets so that the model can be easily train on large dataset with real-time data.

Data Processing:

The raw data that were collected has some missing values and irrelevant columns. After data collection the data was cleaned, handled the missing entries, filtering for Karachi and removing irrelevant information. This ensure data was clean, consistent and ready for analysis and modeling.

Exploratory Data Analysis:

EDA was performed on Jupyter Notebook for easily understanding the results. It was done to understand the pollutant distributions, trends and correlation with AQI. Made some visualizations there such as line charts for checking pollutants over time (Figure 1), a correlation matrix highlighted relationships between different pollutants and AQI (Figure 2), and a bar chart illustrates the distribution of AQI categories to describe air quality and its potential health impact (Figure 3). This analysis helped in identifying important features for model training.

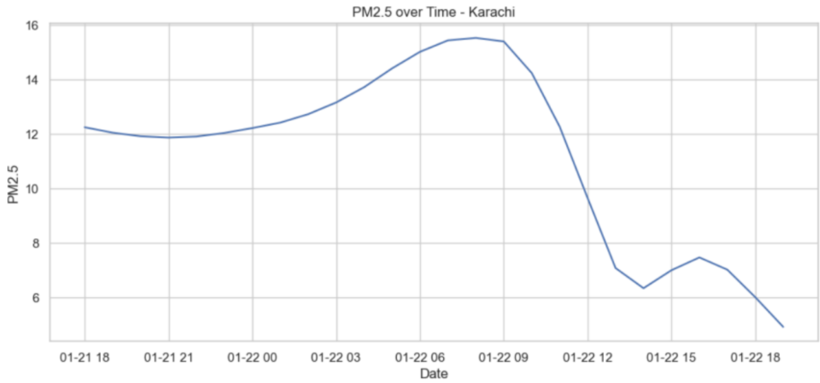

Figure 1: PM2.5 over Time

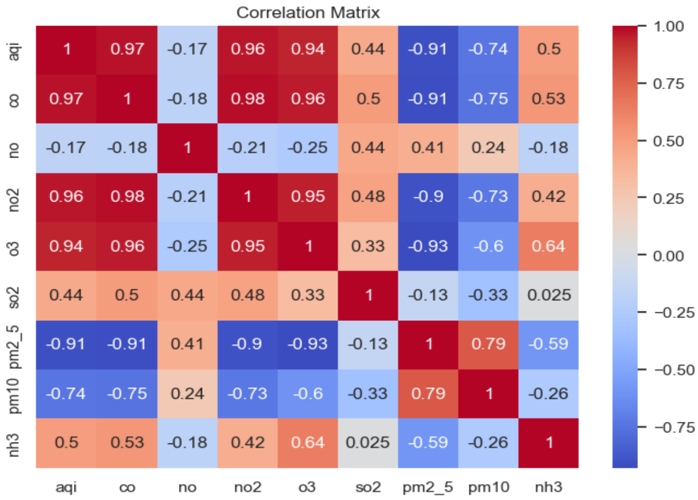

Figure 2: Correlation Matrix

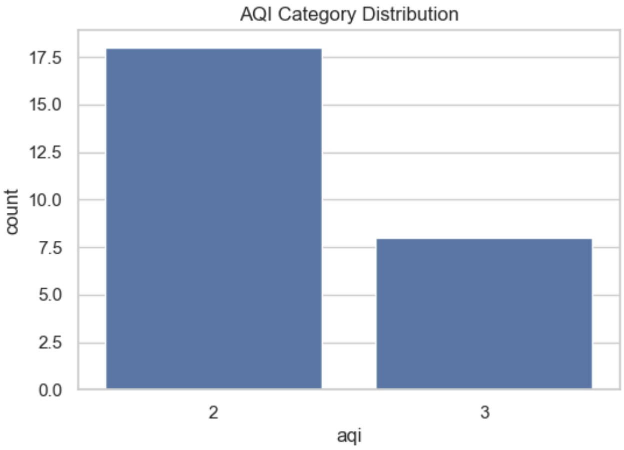

Figure 3: AQI Category

Feature Engineering:

Feature engineering was performed to enhance the performance of model including rolling averages, lag features and weather related metrices from API data. Time based features such as year, month, day of week and weekend indicators were extracted from the date field. The rolling averages were calculated to reduce the fluctuation in the concentrations of pollutants.

After feature engineering, the processed data was stored on feature store (MongoDB) for efficient data management and reuse. Instead of computing features every time, the model pipeline retrieves the preprocessed data directly from database. This helps in improving consistency while reducing redundant computations.

Model Training:

The objective was to train three machine learning models and check which one performs better. The dataset was split into training and validation sets to ensure robust evaluation.

There are many models in machine learning but the for AQI predictions these three models give accurate results and can work on large datasets without crashing. The three models used here are:

- Random Forest Regressor
- XG Boost
- Gradient Boosting

Out of them the best results we get were from Random Forest Regressor.

Model Evaluation:

The models were evaluated based on the values of Mean Absolute Error (MAE) and R^2. The results are in the table below:

Model	MAE	R^2
Random Forest	7.60	0.87
Gradient Boosting	8.29	0.86
XG Boost	8.46	0.85

According to results Random Forest achieved the best performance because it has lowest MAE and highest R^2, which indicates accurate predictions.

Model Deployment:

The project's complete codebase was maintained on GitHub and CI/CD pipeline was implemented using GitActions. It was created for automate testing and deployment workflows. The workflows were scheduled so it ran test every day and updates the predictions automatically for the next 3 days according to the model. This ensures that the updates pushed to the repository were automatically validated and deployed.

Web Application Development:

After completion of model, the entire codebase was pushed to GitHub through CI/CD pipeline. A user-friendly application was developed using Streamlit to visualize AQI predictions for next 3 days with insights. The application was connected directly to GitHub repository which enables easily to get updates through the CI/CD pipeline. This web interface is easily to understand and deployed so anyone with internet can view the AQI trends in Karachi.

Results:

The performance of the trained models was excellent in predicting AQI values for Karachi, and the ensemble-based models performed better in terms of accuracy. Among the models used for prediction, Random Forest had the lowest prediction error and highest explanatory power, which makes it an effective model for dealing with complex relationships between air pollutants, weather, and time variables. Although the performance of the models is satisfactory, it is affected by the availability of historical data, limitations of the API, and the fact that it is limited to a single location.

Conclusion:

This project demonstrated a complete workflow from data collection to model evaluation and deployment of web application for AQI prediction in Karachi. Random Forest result out as the most effective model by accurately forecasting the AQI levels.

Streamlit application link: https://aqipredictorbyaleena.streamlit.app/

GitHub Repository link: https://github.com/Aleenaamir512/aqi_forecast_ml

References:

OpenWeather API: https://openweathermap.org/

AQICN API: https://aqicn.org/station/@544966/

Python Libraries: Numpy, Pandas, XG Boost, Matplotlib, Seaborn, Scikit-learn.